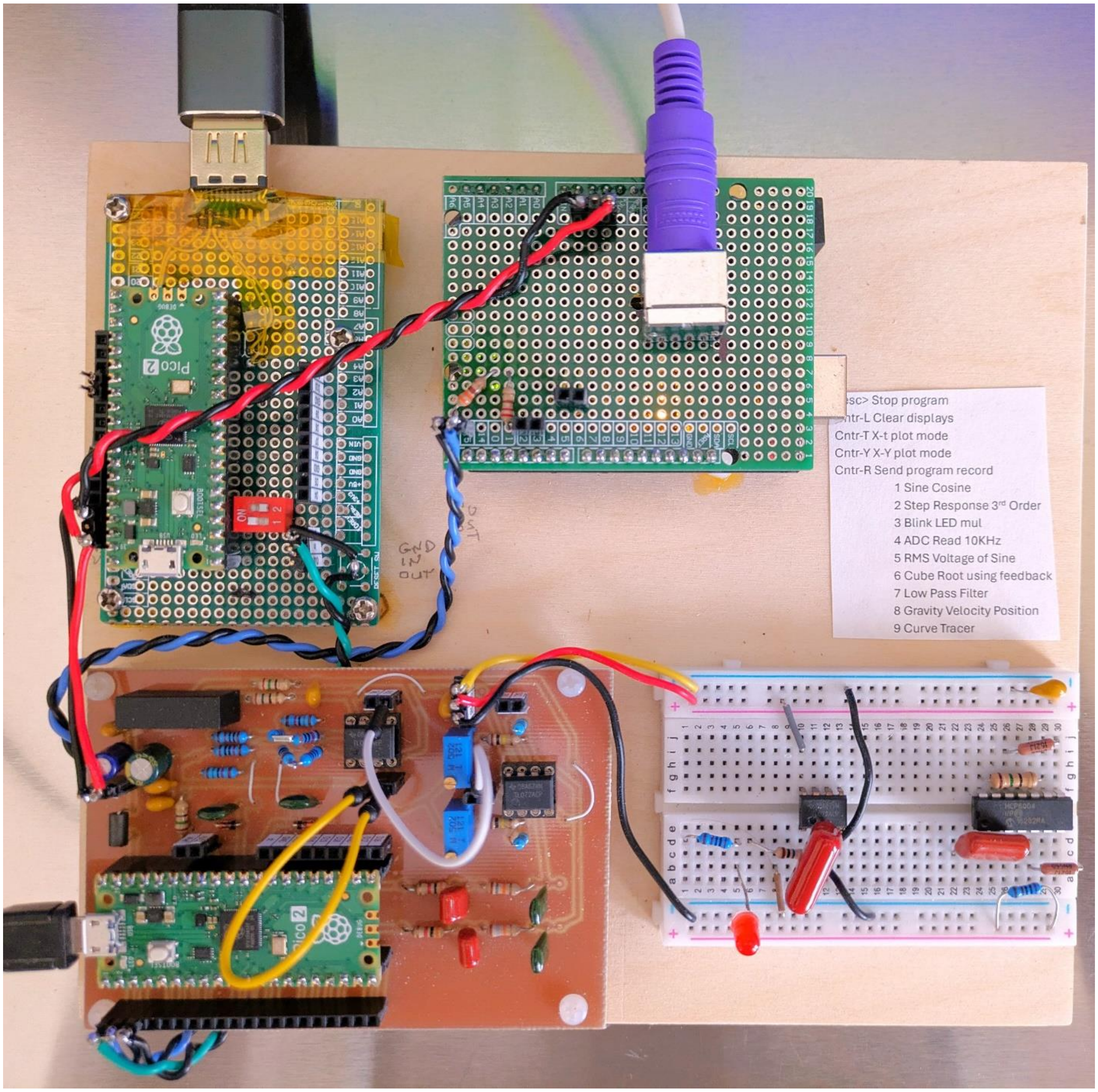
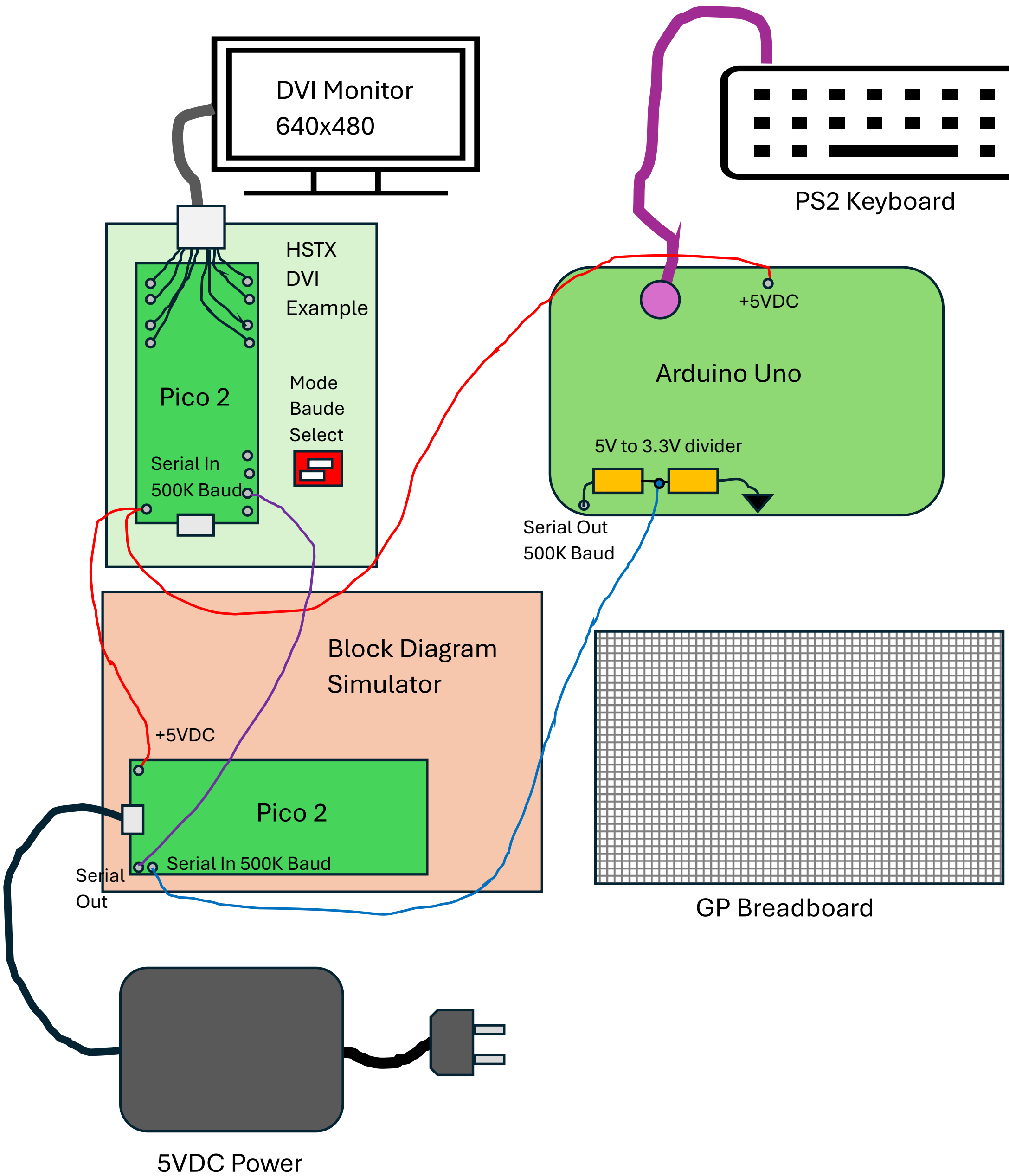
RP2350 HSTX DVI Simple CRT Monitor

(for the Block Simulator) Simple-Circuit 2024

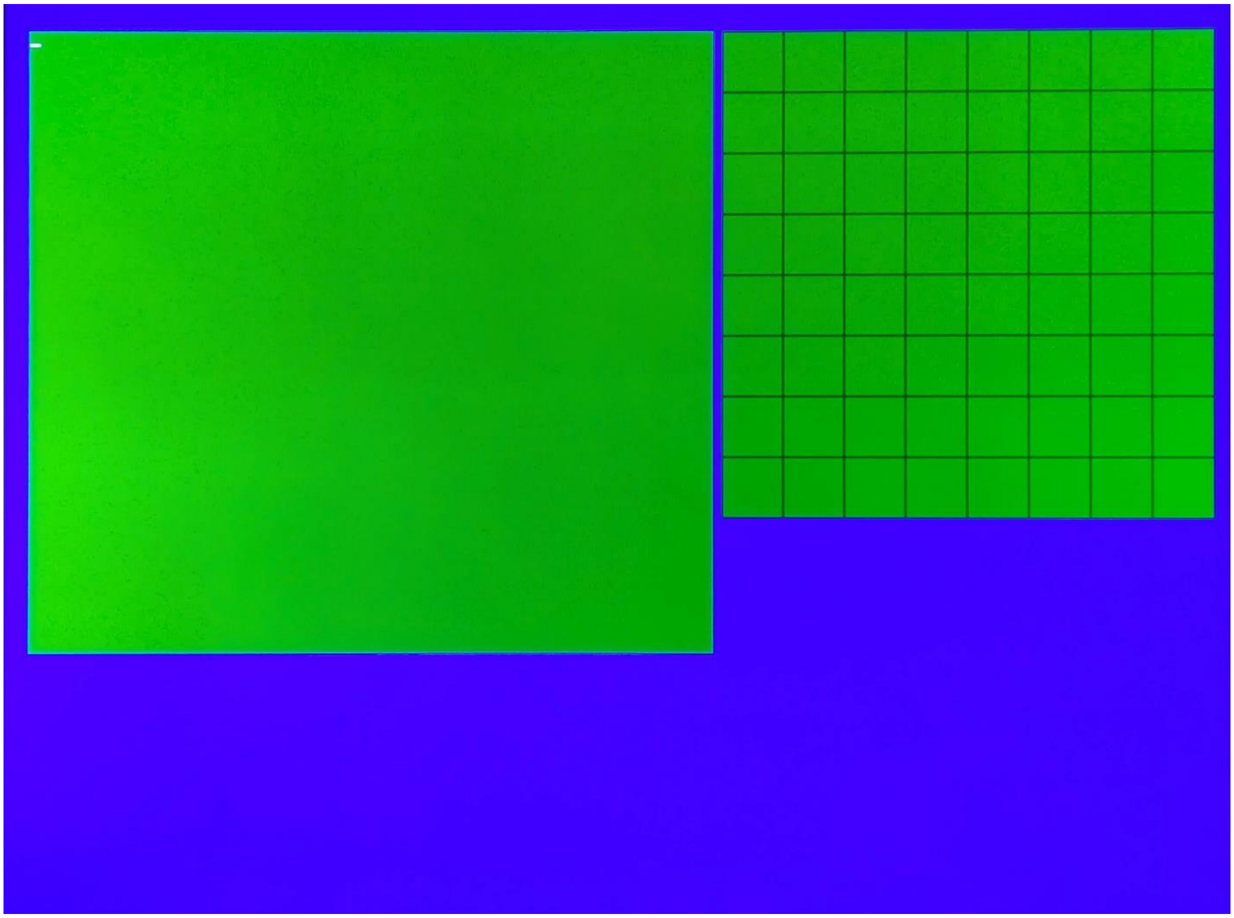


After designing the Analog Computer Block Diagram Simulator, it was time to make it a stand-alone demonstrator. Adding a PS2 keyboard and a DVI video output was all that was needed. The new RP2350 Pico 2 came with an HSTX DVI example program. With a few modifications, it was possible to add text and some graphing capabilities. Example programs could be stored in the keyboard interface reducing the need to type them in.



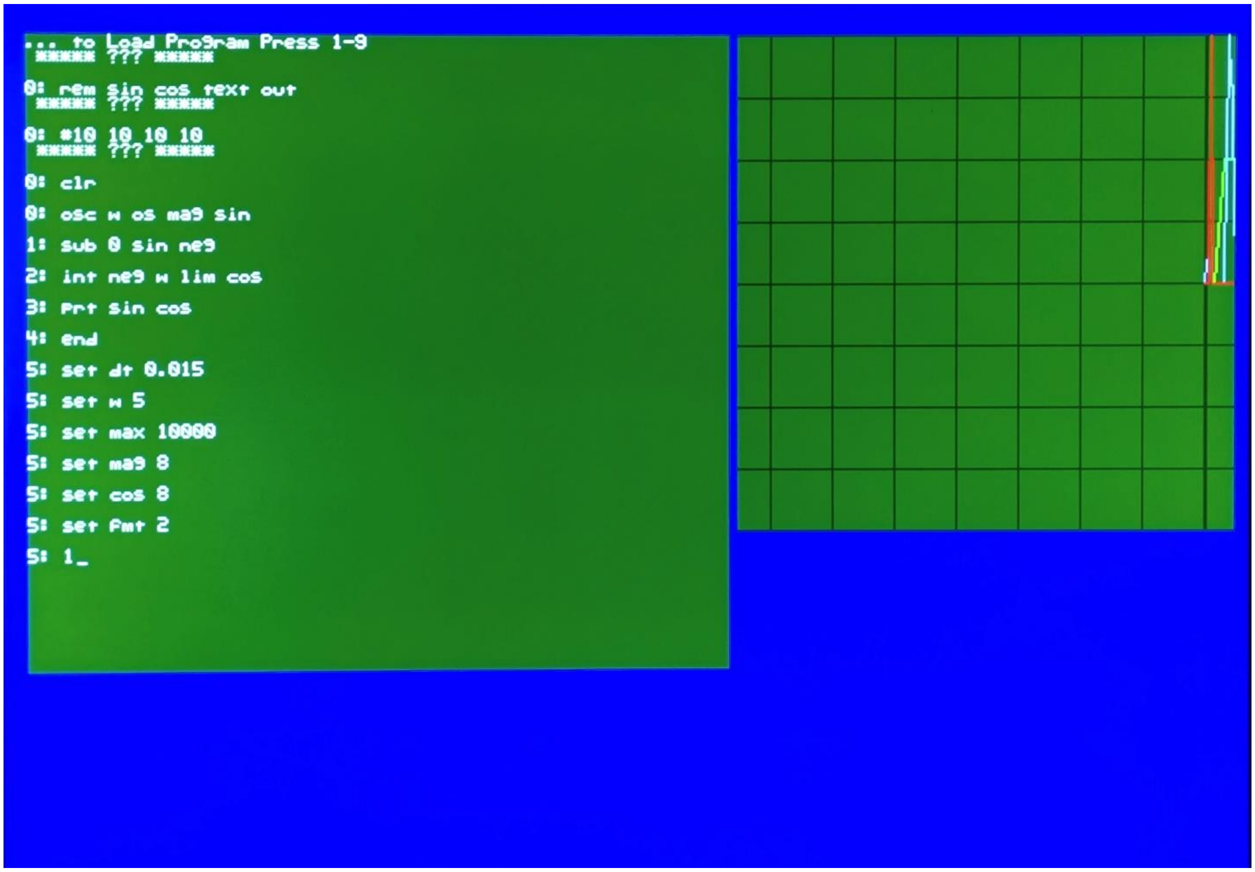
Note: All boards have connected grounds.

The PS2 keyboard input uses a modified Steve Benz's PS2KeyboardHost loaded into an Arduino Uno. His program is reliable and simple to modify. The DVI output is connected per Luke Wren's Pico-DVI-Sock. I hand wired mine, but you may want to purchase one from Adafruit for about $3.



The display program takes data from the UART serial input at 500K baud. There is a select switch to change this to 4800 baud. The green box on the left displays text at 40 lines by 60 characters. Characters use a fixed 5x7 font.

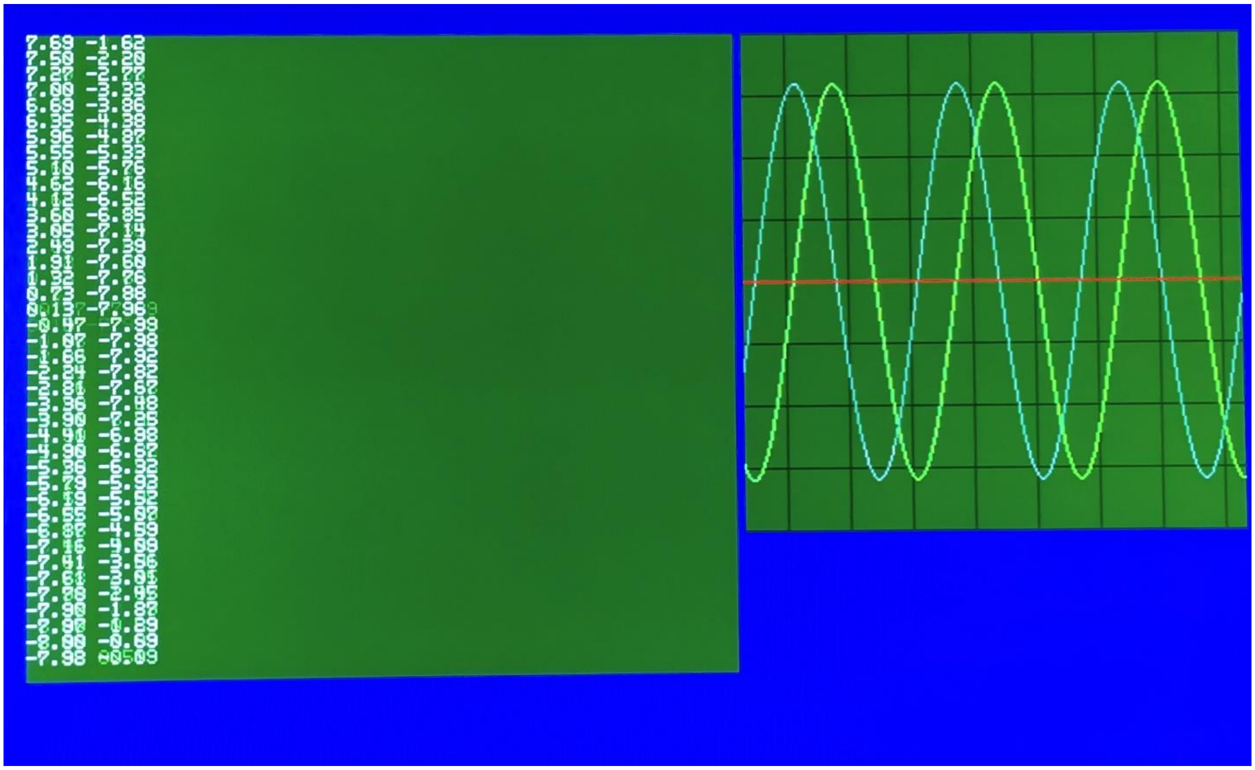
The plot area is 256x256 pixels. Plots are scaled to plus and minus 10. Full scale can be set for up to four plots. Two plot modes are available: x-t strip chart and x-y. Data is any fixed point number separated by a space or comma. Only 0-9, +, and – characters are parsed. Plotting is always on and looks to start if a number starts in the text.



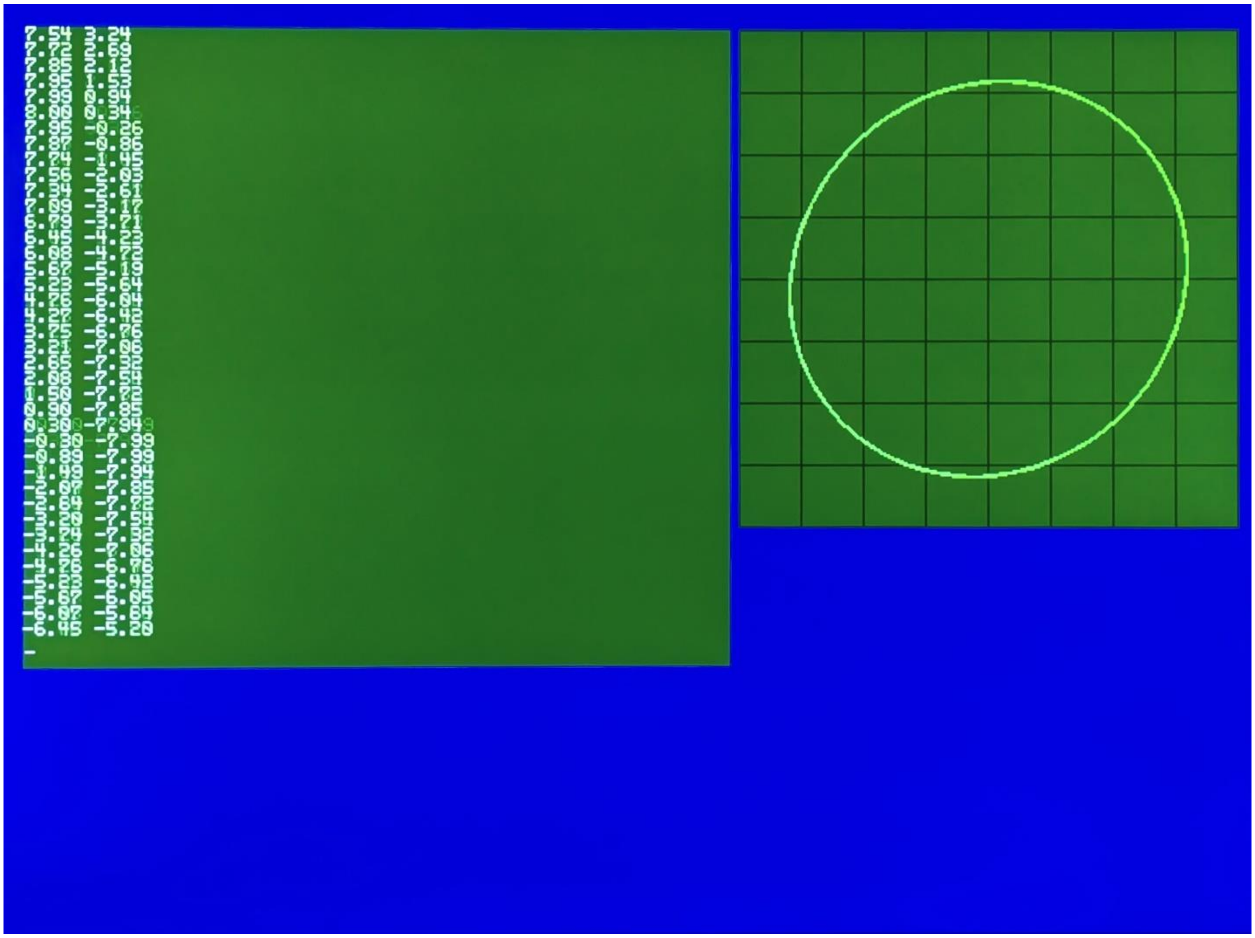
The text display responds to control characters: form feed (cntr-l), carriage return (cntr-m), line feed (cntr-j) and back space (cntr-h).

When cntr-r is pressed, a prompt for example Block Program loading is displayed. Pressing 1 to 9 loads an example program.

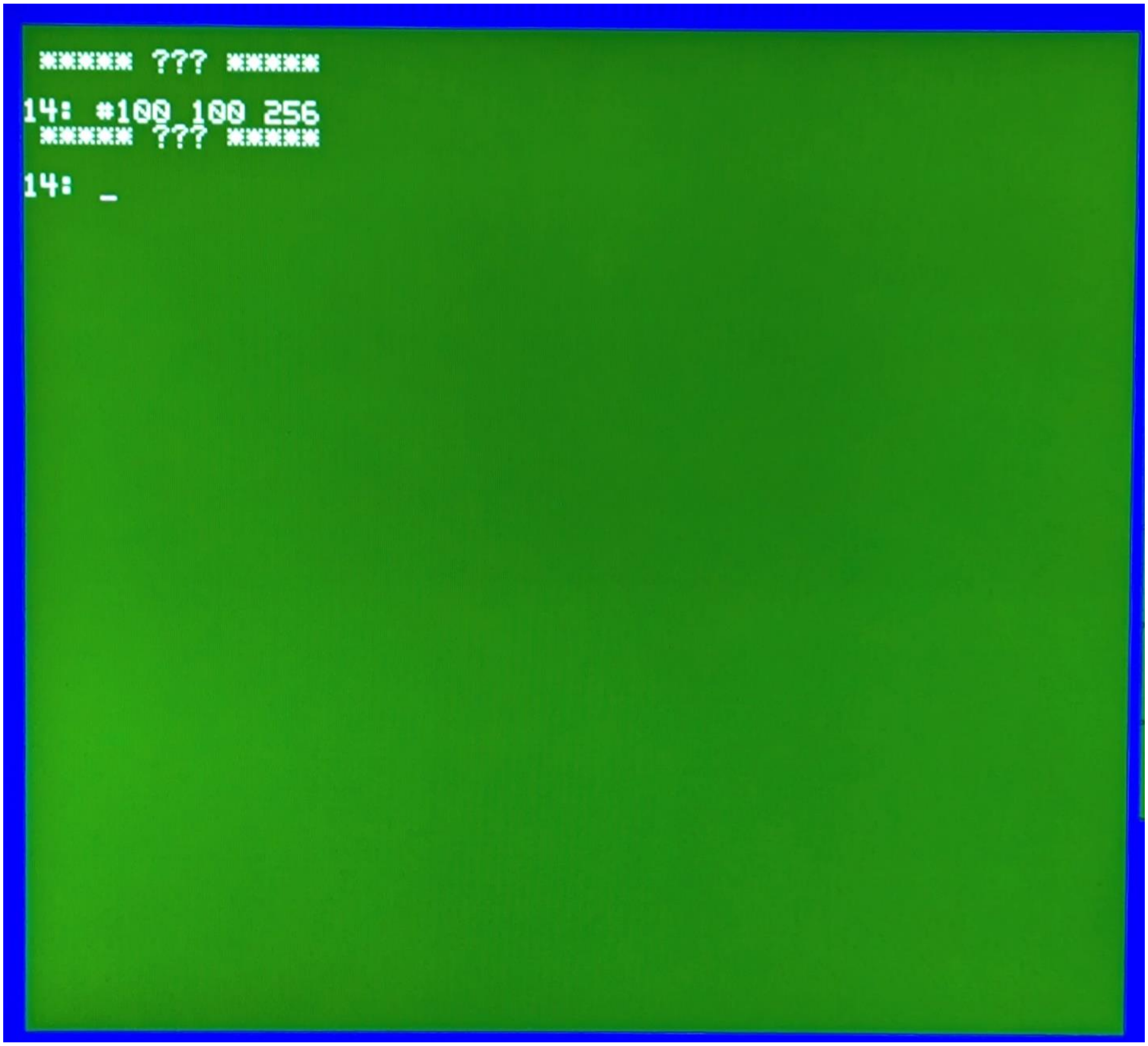
1 - Sine Cosine wave
2 - 3rd order system step response
3 – Blink LED
4 – ADC 10KHz read
5 – Calculate RMS voltage of a sine wave
6 – Find the cube root using negative feedback
7 – Low Pass Filter
8 – Integrate to find Velocity and Position from Acceleration
9 – A simple curve tracer



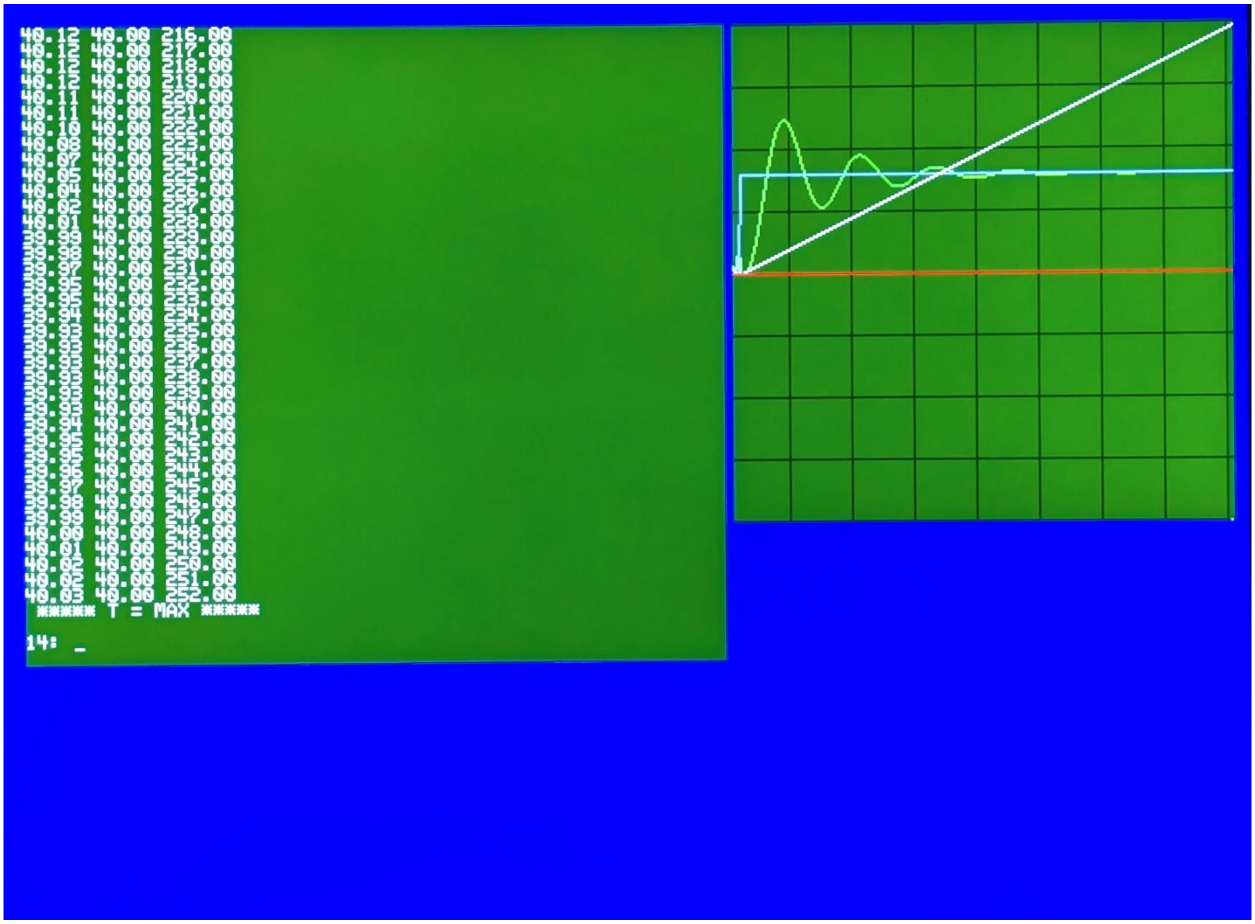
Example of x-t ploting for the sine cosine example. Pressing cntr-t selects the x-t mode. To clear the displays, press cntr-l.



The x-y mode is set by pressing cntr-y. The first numeric value is the x-axis and the second numeric value is the y-axis. If more than two number columns are present, only the first two are used.



Plot full scale is set by typing # followed by one to four space delimited numbers. #100 100 256<cr>
Sets plot 1 to ±100, plot 2 to ±100 and plot 3 to ±256. Zero is always mid scale.



The program 2 example presets the scales when loading. When the program runs: the step input, the output response and the time are plotted.

Resources for building this project:

https://github.com/simple-circuit/RP2350-Dumb-Terminal

- PS2KeyboardHost2_b Modified PS2 Keyboard example for Uno processor
- Pico_block_compiler_p Modified Block diagram compliers for UART in/out
- pico_hstx_example_term Modified dvi_out_hstx_encoder example

https://github.com/simple-circuit/Block-Diagram-Compiler

https://github.com/Wren6991/Pico-DVI-Sock

https://github.com/raspberrypi/pico-examples/tree/master/hstx/dvi_out_hstx_encoder

https://github.com/SteveBenz/PS2KeyboardHost

https://www.adafruit.com/product/5957

Arduino UNO PS2 Keyboard Wiring